

Module 08 - Balances & Settlement

Net balances and simplified settlement suggestions

SOURCE OF TRUTH: This resource intentionally uses plain JavaScript. The locked stack is React + JavaScript + Firebase + Material UI (MUI). Do not introduce Next.js, TypeScript, shadcn/ui, Redux, or another backend framework unless the project owner explicitly changes this specification.

Scope

Convert expense/split records into net user balances for a group or plan and provide a simplified set of suggested payments. V1 can support manual "mark settled"; actual money transfer is out of scope.

Concept

For each expense:
payer receives +amount
each participant owes -their share

Net balance > 0 => user should receive
Net balance < 0 => user should pay

Example

Rahul owes Ved 500
Aman owes Ved 500
Ved owes Priya 300

Suggested simplified settlement:
Rahul -> Ved 500
Aman -> Ved 200
Aman -> Priya 300

Algorithm requirements

- Aggregate net balances per user.
- Separate creditors and debtors.
- Match largest debtor with largest creditor iteratively.
- Round only at the final minor-unit level.
- Do not lose traceability: suggested settlement is derived from underlying expenses.

Settlement record

```
groups/{groupId}/settlements/{settlementId}
{
  fromUserId,
  toUserId,
  amount,
  currency,
  relatedExpenseIds,
  createdBy,
  createdAt,
  settledAt,
  status: "suggested" | "confirmed" | "cancelled"
}
```

Acceptance criteria

- Net balances reconcile to zero across a closed set of expenses.
- Suggested payments never create money out of thin air.

- Users can mark a settlement confirmed without payment processing.
- Historical expenses remain unchanged when a settlement is recorded.